

Kaustubh Rathi

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SUMMARY

Machine Learning Engineer | Applied ML / Research. ML engineer with from-scratch deep-learning implementations (Transformers, attention, CNN-RNN, neural recommenders) and production LLM-agent work at Amazon. Codeforces Expert and LeetCode Knight with strong algorithms, math, and systems fundamentals.

EDUCATION

B.Tech in Information Technology (Minor in Economics) 2022 – 2026
Indian Institute of Information Technology (IIIT), Allahabad — CGPA: 8.52/10 Prayagraj, India

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Generative AI, Data Mining

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, TensorFlow, Transformers, attention, seq2seq, CNNs, embeddings, RAG, NER, recommender systems, scikit-learn

LLM & Agents: LLM agents, Amazon Bedrock, tool use, RAG, knowledge bases

MLOps & Cloud: AWS (SageMaker, Bedrock, Lambda), Docker, Git

Languages: Python, C++, Java, SQL

EXPERIENCE

Software Development Engineer Intern Jan 2026 – Present
Amazon — Grocery Ordering, Stores Bengaluru, India

- Architected an autonomous order-failure remediation agent on Amazon Bedrock: a multi-stage LLM pipeline that diagnoses ~150 failed orders/day, semantically clusters them by root cause, and emits confidence-scored mitigations with evidence, auto-generating code-fix CRs to prevent recurrence.
- Designed the agent's reliability layer (per-call output validation, automatic retries, 1–5 confidence gating, human approval gates) for trustworthy production decisioning.
- Shipped a 24/7 LLM incident-investigation agent (Bedrock + tool use) end to end that correlates multi-source telemetry with deployments to auto-generate root-cause analyses, cutting investigation time from ~2 hours to ~15 minutes.

Artificial Intelligence Intern Dec 2024 – Feb 2025
ASPR Agrovalue (I) Limited India

- Automated an NER-based invoice-extraction pipeline pulling structured fields from 10,000+ supplier invoices/month into a normalized schema, replacing manual data entry.
- Developed a Python web-scraping + ETL pipeline (1,000+ price records/day) and a Streamlit dashboard the procurement team adopted for price-trend visibility.

PROJECTS

Transformer & Attention Seq2Seq (from scratch) | PyTorch, NMT, BLEU, TensorBoard 🔗 Transformer, 🔗 Attention

- Implemented the Transformer from scratch in PyTorch (512-dim model, 8 attention heads, 3+3 encoder/decoder layers) for German-English machine translation on Multi30k, achieving BLEU-4 of 29.4; evaluated with a full BLEU harness and TensorBoard loss/LR tracking.
- Prototyped a Bahdanau-style attention encoder-decoder LSTM baseline for the same task with teacher forcing, demonstrating core seq2seq fundamentals.

Image Captioning (CNN-RNN) | PyTorch, torchvision, LSTM 🔗 GitHub

- Trained a CNN-encoder (torchvision ResNet-50) + LSTM-decoder captioning model on Flickr8k (8k images, 40k captions), achieving BLEU-4 of 22.1 on the validation split with a custom vocabulary of 8,000 tokens.

Neural Collaborative Filtering | PyTorch Lightning, MovieLens 🔗 GitHub

- Modeled a recommender learning 50-dim user/item embeddings on MovieLens-1M (1M+ ratings) with PyTorch Lightning, reaching 0.79 MAE on a held-out 20% split — ~16% better than a global-mean baseline (0.94).

Deep Researcher | Python, LangGraph, FastAPI, Flask, WebSocket, FAISS 🔗 GitHub

- Orchestrated a multi-agent research-automation web app with 7 specialized LLM agents, a FastAPI backend, and WebSocket log streaming; exports reports to PDF/DOCX/Markdown.

ACHIEVEMENTS

- **Codeforces Expert** (max rating 1751) | **LeetCode Knight** (rating 2000+), 800+ problems solved.
- Secured **99.45 percentile** in JEE Mains 2022 (top 0.55% among 1M+ candidates).
- **Overall Coordinator, Robita Club (Generative AI), IIIT Allahabad** (2023–25): led 10+ students, mentored 100+ juniors via AI hackathons and PyTorch workshops.